Topic Name:
Search Strategies

Search Algorithms

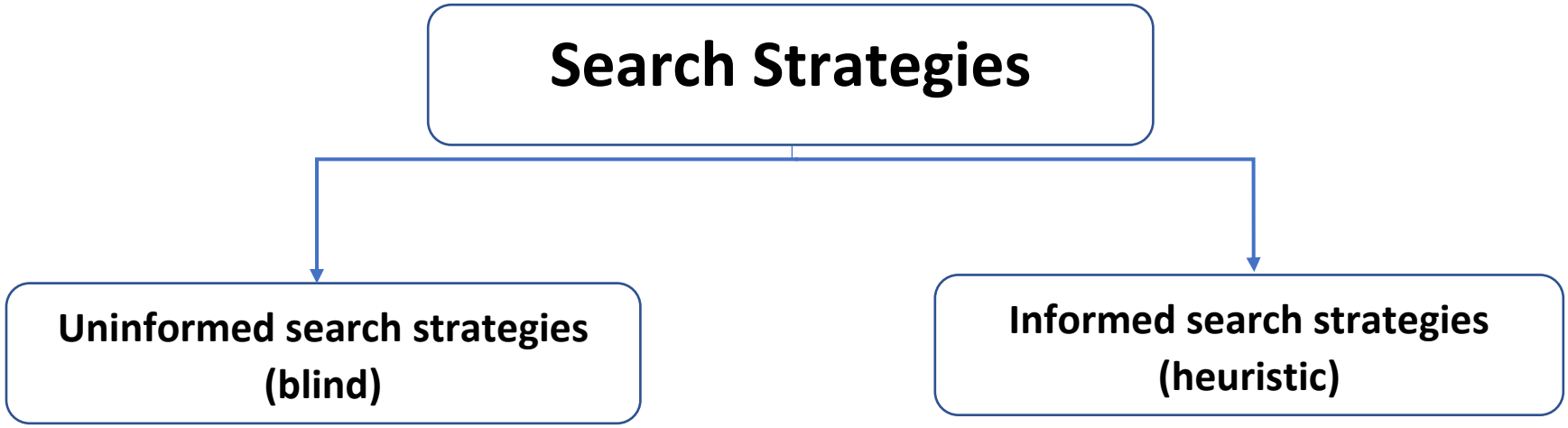
Problem -----> Solution

Solution ⟹ Search strategy to find the goal

Measuring problem-solving performance

- **Completeness**: Is the algorithm guaranteed to find a solution when there is one?
- **Optimality**: Does the strategy find the optimal solution?
- **Time complexity**: How long does it take to find a solution?
- **Space complexity**: How much memory is needed to perform the search?

Search Strategies



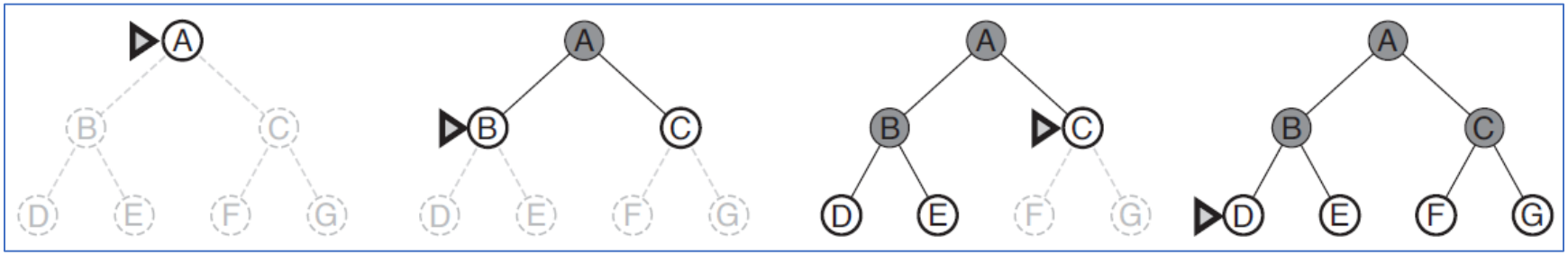
Uninformed search strategies

The term means that the strategies have no additional information about states beyond that provided in the problem definition.

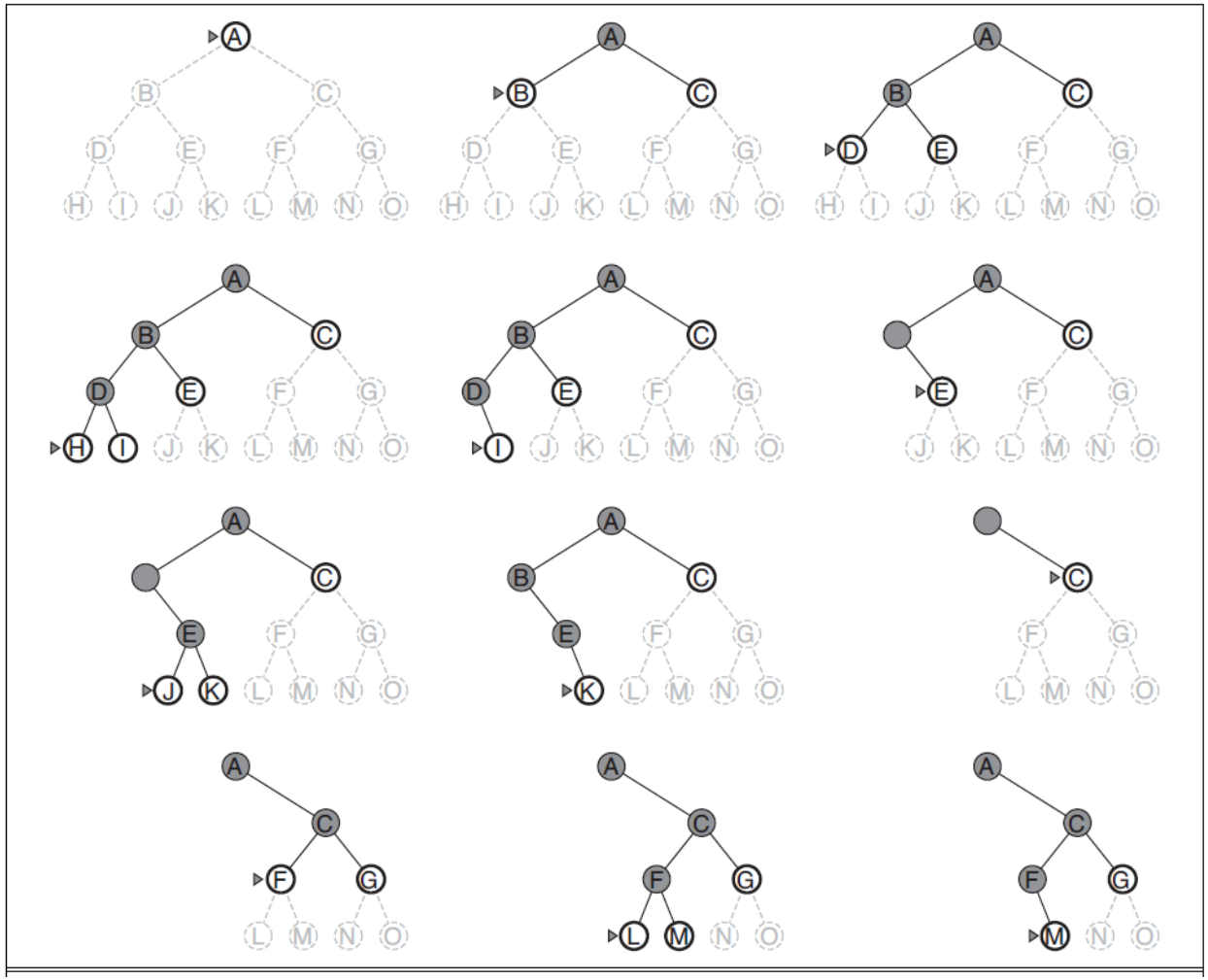
Examples of Uninformed search strategies: Breadth-first search and Depth-first search

All blind search strategies are distinguished by the order in which nodes are expanded.

Breadth-first search



Depth-first search

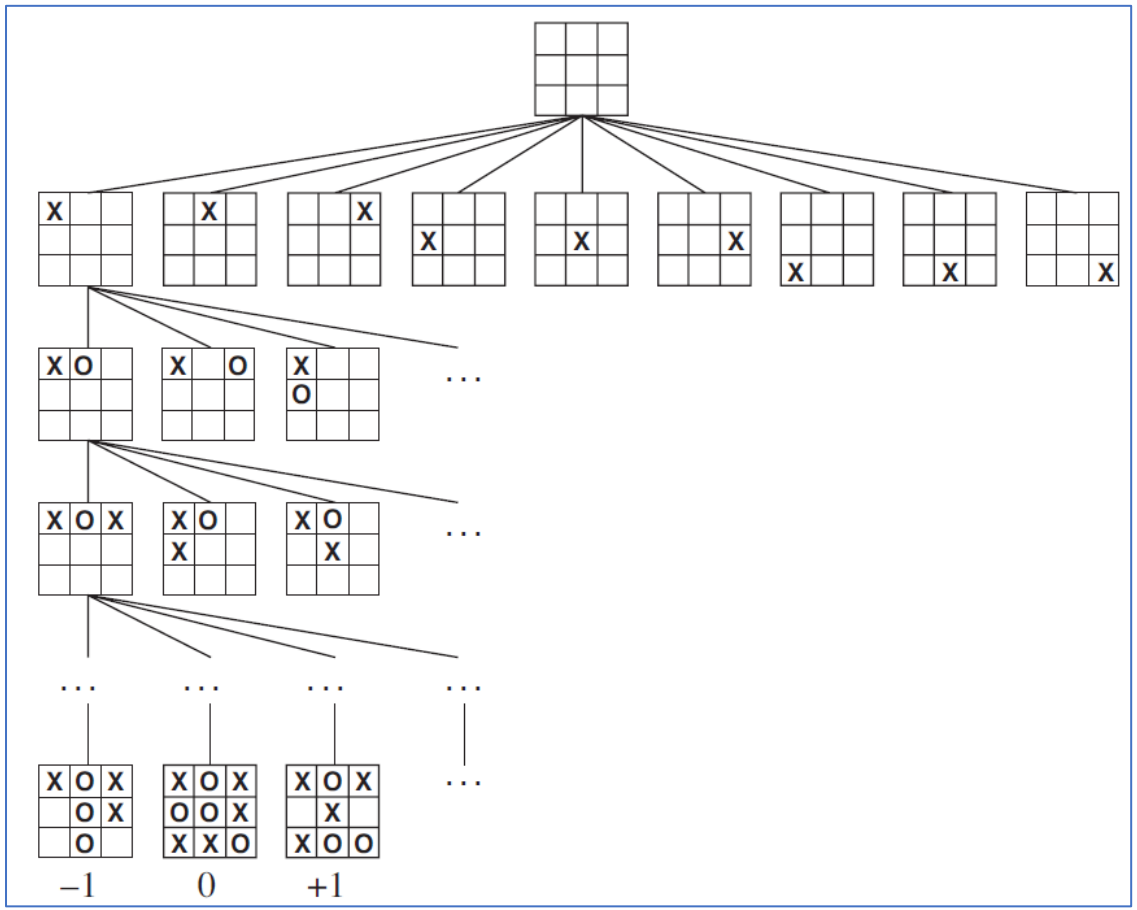


Informed search strategies

One that uses problem-specific knowledge beyond the definition of the problem itself and then it can find solutions more efficiently than can an uninformed strategy. Strategies that know whether one non-goal state is “more promising” than another are called informed search.

Heuristic function, denoted h(n):
h(n) = estimated cost of the cheapest path from the state at node n to a goal state.

Brute-force or Optimization?



Two common problems

N-puzzle problem

Example of 8-puzzle.

5		2
8	4	7
6	1	3

Initial state

	1	2
3	4	5
6	7	8

Goal state

Example of 15-puzzle

13	2	4	5
1	3	14	7
8		10	12
11	9	6	15

Initial state

	1	2	3
4	5	6	7
8	9	10	11
12	13	14	15

Goal state

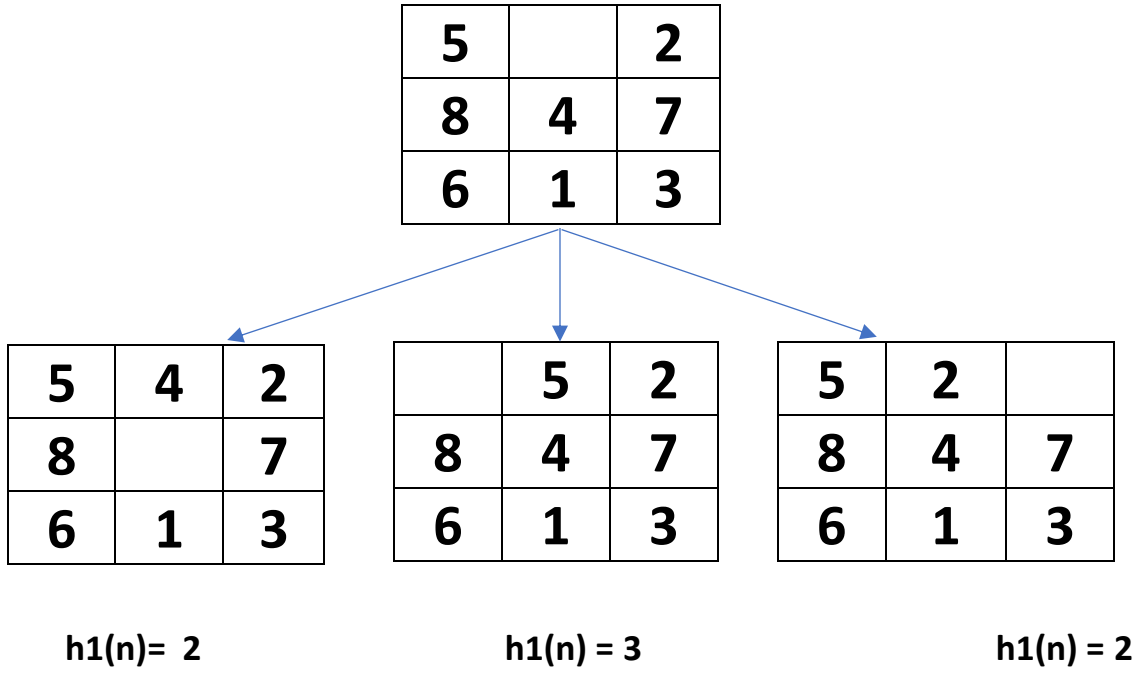
Heuristic function

h1(n): # of tiles that are correctly placed

h2(n): # of tiles that are incorrectly placed

h3(n): sum of distances.

***** Cost and objective function**


5 2
8 4 7
6 1 3
5 4 2
8 7
6 1 3
5 2
8 4 7
6 1 3
5 2
8 4 7
6 1 3
h1(n)= 2
h1(n) = 3
h1(n) = 2

N- queens problem

Example of 4-queens problem.

		*	
*			
			*
	*		

Heuristic function

h1(n): is the number of pairs of queens that are attacking each other
h2(n): the number of nonattacking pairs of queens

	*			
*				
		*	*	
				*